

Alterations in immunized antigens of *Anisakis pegreffii* by ampicillin-induced gut microbiome changes in mice.

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The gut microbiome plays an essential role in host immune responses, including allergic reactions. However, commensal gut microbiota is extremely sensitive to antibiotics and excessive usage can cause microbial dysbiosis. Herein, we investigated how changes in the gut microbiome induced by ampicillin affected the production of IgG1 and IgG2a antibodies in mice subsequently exposed to *Anisakis pegreffii* antigens. Ampicillin treatment caused a notable change in the gut microbiome as shown by changes in both alpha and beta diversity indexes. In a 1-dimensional immunoblot using *Anisakis*-specific anti-mouse IgG1, a 56-kDa band corresponding to an unnamed *Anisakis* protein was detected using mass spectrometry analysis only in ampicillin-treated mice. In the *Anisakis*-specific anti-mouse IgG2a-probed immunoblot, a 70-kDa band corresponding to heat shock protein 70 (HSP70) was only detected in ampicillin-treated and *Anisakis*-immunized mice. A 2-dimensional immunoblot against *Anisakis* extract with immunized mouse sera demonstrated altered spot patterns in both groups. Our results showed that ampicillin treatment altered the gut microbiome composition in mice, changing the immunization response to antigens from *A. pegreffii*. This research could serve as a basis for developing vaccines or allergy immunotherapies against parasitic infections.

Different Dynamic Patterns of β -Lactams, Quinolones, Glycopeptides and Macrolides on Mouse Gut Microbial Diversity.

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The adverse impact of antibiotics on the gut microbiota has attracted extensive interest, particularly due to the development of microbiome research techniques in recent years. However, a direct comparison of the dynamic effects of various types of antibiotics using the same animal model has not been available. In the present study, we selected six antibiotics from four categories with the broadest clinical usage, namely, β -lactams (Ceftriaxone Sodium, Cefoperazone/Sulbactam and meropenem), quinolones (ofloxacin), glycopeptides (vancomycin), and macrolides (azithromycin), to treat BALB/c mice. Stool samples were collected during and after the administration of antibiotics, and microbial diversity was analyzed through Illumina sequencing and bioinformatics analyses using QIIME. Both α and β diversity analyses showed that ceftriaxone sodium, cefoperazone/sulbactam, meropenem and vancomycin changed the gut microbiota dramatically by the second day of antibiotic administration whereas the influence of ofloxacin was trivial. Azithromycin clearly changed the gut microbiota but much less than vancomycin and the β -lactams. In general, the community changes induced by the three β -lactam antibiotics showed consistency in inhibiting *Papillibacter*, *Prevotella* and *Alistipes* while inducing massive growth of *Clostridium*. The low diversity and high *Clostridium* level might be an important cause of *Clostridium difficile* infection after usage of β -lactams. Vancomycin was unique in that it inhibited Firmicutes, mainly the genus *Clostridium*. On the other hand, it induced the growth of *Escherichia* and effect lasted for months afterward. Azithromycin and meropenem induced the growth of *Enterococcus*. These findings will be useful for understanding the potential adverse effects of antibiotics on the gut microbiome and ensuring their better usage.

Rare variant associations with plasma protein levels in the UK Biobank.

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Integrating human genomics and proteomics can help elucidate disease mechanisms, identify clinical biomarkers and discover drug targets¹⁻⁴. Because previous proteogenomic studies have focused on common variation via genome-wide association studies, the contribution of rare variants to the plasma proteome remains largely unknown. Here we identify associations between rare protein-coding variants and 2,923 plasma protein abundances measured in 49,736 UK Biobank individuals. Our variant-level exome-wide association study identified 5,433 rare genotype-protein associations, of which 81% were undetected in a previous genome-wide association study of the same cohort⁵. We then looked at aggregate signals using gene-level collapsing analysis, which revealed 1,962 gene-protein associations. Of the 691 gene-level signals from protein-truncating variants, 99.4% were associated with decreased protein levels. STAB1 and STAB2, encoding scavenger receptors involved in plasma protein clearance, emerged as pleiotropic loci, with 77 and 41 protein associations, respectively. We demonstrate the utility of our publicly accessible resource through several applications. These include detailing an allelic series in NLRC4, identifying potential biomarkers for a fatty liver disease-associated variant in HSD17B13 and bolstering phenome-wide association studies by integrating protein quantitative trait loci with protein-truncating variants in collapsing analyses. Finally, we uncover distinct proteomic consequences of clonal haematopoiesis (CH), including an association between TET2-CH and increased FLT3 levels. Our results highlight a considerable role for rare variation in plasma protein abundance and the value of proteogenomics in therapeutic discovery.

Macrophages in Recurrent Glioblastoma as a Prognostic Factor in the Synergistic System of the Tumor Microenvironment.

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Glioblastoma (GBM) is a common and highly malignant primary tumor of the central nervous system in adults. Ever more recent papers are focusing on understanding the role of the tumor microenvironment (TME) in affecting tumorigenesis and the subsequent prognosis. We assessed the impact of macrophages in the TME on the prognosis in patients with recurrent GBM. A PubMed, MEDLINE and Scopus review was conducted to identify all studies dealing with macrophages in the GBM microenvironment from January 2016 to December 2022. Glioma-associated macrophages (GAMs) act critically in enhancing tumor progression and can alter drug resistance, promoting resistance to radiotherapy and establishing an immunosuppressive environment. M1 macrophages are characterized by increased secretion of proinflammatory cytokines, such as IL-1 β , tumor necrosis factor (TNF), IL-27, matrix metalloproteinase (MMPs), CCL2, and VEGF (vascular endothelial growth factor), IGF1, that can lead to the destruction of the tissue. In contrast, M2 is supposed to participate in immunosuppression and tumor progression, which is formed after being exposed to the macrophage M-CSF, IL-10, IL-35 and the transforming growth factor- β (TGF- β). Because there is currently no standard of care in recurrent GBM, novel identified targeted therapies based on the complex signaling and interactions between the glioma stem cells (GSCs) and the TME, especially resident microglia and bone-marrow-derived macrophages, may be helpful in improving the overall survival of these patients in the near future.

Single-Cell Transcriptomics of Glioblastoma Reveals a Unique Tumor Microenvironment and Potential Immunotherapeutic Target Against Tumor-Associated Macrophage.

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The main immune cells in GBM are tumor-associated macrophages (TAMs). Thus far, the studies investigating the activation status of TAM in GBM are mainly limited to bulk RNA analyses of individual tumor biopsies. The activation states and transcriptional signatures of TAMs in GBM remain poorly characterized. We comprehensively analyzed single-cell RNA-sequencing data, covering a total of 16,201 cells, to clarify the relative proportions of the immune cells infiltrating GBMs. The origin and TAM states in GBM were characterized using the expression profiles of differential marker genes. The vital transcription factors were examined by SCENIC analysis. By comparing the variable gene expression patterns in different clusters and cell types, we identified components and characteristics of TAMs unique to each GBM subtype. Meanwhile, we interrogated the correlation between SPI1 expression and macrophage infiltration in the TCGA-GBM dataset. The expression patterns of TMEM119 and MHC-II can be utilized to distinguish the origin and activation states of TAMs. In TCGA-Mixed tumors, almost all TAMs were bone marrow-derived macrophages. The TAMs in TCGA-proneural tumors were characterized by primed microglia. A different composition was observed in TCGA-classical tumors, which were infiltrated by repressed microglia. Our results further identified SPI1 as a crucial regulon and potential immunotherapeutic target important for TAM maturation and polarization in GBM. We describe the immune landscape of human GBM at a single-cell level and define a novel categorization scheme for TAMs in GBM. The immunotherapy against SPI1 would reprogram the immune environment of GBM and enhance the treatment effect of conventional chemotherapy drugs.

Glioblastoma-instructed microglia transition to heterogeneous phenotypic states with phagocytic and dendritic cell-like features in patient tumors and patient-derived orthotopic xenografts.

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A major contributing factor to glioblastoma (GBM) development and progression is its ability to evade the immune system by creating an immune-suppressive environment, where GBM-associated myeloid cells, including resident microglia and peripheral monocyte-derived macrophages, play critical pro-tumoral roles. However, it is unclear whether recruited myeloid cells are phenotypically and functionally identical in GBM patients and whether this heterogeneity is recapitulated in patient-derived orthotopic xenografts (PDOXs). A thorough understanding of the GBM ecosystem and its recapitulation in preclinical models is currently missing, leading to inaccurate results and failures of clinical trials. Here, we report systematic characterization of the tumor microenvironment (TME) in GBM PDOXs and patient tumors at the single-cell and spatial levels. We applied single-cell RNA-sequencing, spatial transcriptomics, multicolor flow cytometry, immunohistochemistry and functional studies to examine the heterogeneous TME instructed by GBM cells. GBM PDOXs representing different tumor phenotypes were compared to glioma mouse GL261 syngeneic model and patient tumors. We show that GBM tumor cells reciprocally interact with host cells to create a GBM patient-specific TME in PDOXs. We detected the most prominent transcriptomic adaptations in myeloid cells, with brain-resident microglia representing the main population in the cellular tumor, while peripheral-derived myeloid cells infiltrated the brain at sites of blood-brain barrier disruption. More specifically, we show that GBM-educated microglia undergo transition to diverse phenotypic states across distinct GBM landscapes and tumor niches. GBM-educated microglia

subsets display phagocytic and dendritic cell-like gene expression programs. Additionally, we found novel microglial states expressing cell cycle programs, astrocytic or endothelial markers. Lastly, we show that temozolomide treatment leads to transcriptomic plasticity and altered crosstalk between GBM tumor cells and adjacent TME components. Our data provide novel insights into the phenotypic adaptation of the heterogeneous TME instructed by GBM tumors. We show the key role of microglial phenotypic states in supporting GBM tumor growth and response to treatment. Our data place PDOXs as relevant models to assess the functionality of the TME and changes in the GBM ecosystem upon treatment.

Specialized functions and sexual dimorphism explain the functional diversity of the myeloid populations during glioma progression.

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Malignant gliomas are aggressive, hard-to-treat brain tumors. Their tumor microenvironment is massively infiltrated by myeloid cells, mostly brain-resident microglia, bone marrow (BM)-derived monocytes/macrophages, and dendritic cells that support tumor progression. Single-cell omics studies significantly dissected immune cell heterogeneity, but dynamics and specific functions of individual subpopulations were poorly recognized. We use Cellular Indexing of Transcriptomes and Epitopes by sequencing (CITE-seq) to precisely dissect myeloid cell identities and functionalities in murine GL261 gliomas. We demonstrate that the diversity of myeloid cells infiltrating gliomas is dictated by cell type and cell state. Glioma-activated microglia are the major source of cytokines attracting other immune cells, whereas BM-derived cells show the monocyte-to-macrophage transition in the glioma microenvironment. This transition is coupled with a phenotypic switch from the IFN-related to antigen-presentation and tumor-supportive gene expression. Moreover, we found sex-dependent differences in transcriptional programs and composition of myeloid cells in murine and human glioblastomas.

M2-polarized macrophages contribute to neovasculogenesis, leading to relapse of oral cancer following radiation.

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Despite the fact that radiation is one of the standard therapies in the treatment of patients with oral cancer, tumours can recur even in the early stages of the disease, negatively impacting prognosis and quality of life. We previously found that CD11b(+) bone marrow-derived cells (BMDCs) were recruited into human glioblastoma multiforme (GBM), leading to re-organization of the vasculature and tumour regrowth. However, it is not yet known how these cells contribute to tumour vascularization. In the present study, we investigated the role of infiltrating CD11b(+) myeloid cells in the vascularization and recurrence of oral squamous cell carcinoma (OSCC). In a xenograft mouse model, local irradiation caused vascular damage and hypoxia in the tumour and increased infiltration of CD11b(+) myeloid cells. These infiltrating cells showed characteristics of M2 macrophages (M2Mφs) and are associated with the promotion of vascularization. M2Mφs promoted tumour progression in recurrence after irradiation compared to non-irradiated tumours. In addition, we found that CD11b(+) myeloid cells, as well as CD206(+) M2Mφs, are increased during recurrence after radiotherapy in human OSCC specimens. Our findings may lead to the development of potential clinical biomarkers or treatment targets in irradiated

OSCC patients.

Intravital 2-photon imaging reveals distinct morphology and infiltrative properties of glioblastoma-associated macrophages.

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Characterized by a dismal survival rate and limited response to therapy, glioblastoma (GBM) remains one of the most aggressive human malignancies. Recent studies of the role of tumor-associated macrophages (TAMs) in the progression of GBMs have demonstrated that TAMs are significant contributors to tumor growth, invasion, and therapeutic resistance. TAMs, which include brain-resident microglia and circulating bone marrow derived-monocytes (BMDMs), are typically grouped together in histopathological and molecular analyses due to the lack of reliable markers of distinction. To develop more effective therapies aimed at specific TAM populations, we must first understand how these cells differ both morphologically and behaviorally. Furthermore, we must develop a deeper understanding of the mechanisms encouraging their infiltration and how these mechanisms can be therapeutically exploited. In this study, we combined immunocompetent lineage tracing mouse models of GBM with high-resolution open-skull 2-photon microscopy to investigate the phenotypical and functional characteristics of TAMs. We demonstrate that TAMs are composed of 2 morphologically distinct cell types that have differential migratory propensities. We show that BMDMs are smaller, minimally branched cells that are highly migratory compared with microglia, which are larger, highly branched stationary cells. In addition, 2 populations of monocytic macrophages were observed that differed in terms of CX3CR1 expression and migratory capacity. Finally, we demonstrate the efficacy of anti-vascular endothelial growth factor A blockade for prohibiting TAM infiltration, especially against BMDMs. Taken together, our data clearly define characteristics of individual TAM populations and suggest that combination therapy with antivascular and antichemotaxis therapy may be an attractive option for treating these tumors.

TNF α secreted by glioma associated macrophages promotes endothelial activation and resistance against anti-angiogenic therapy.

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One of the most prominent features of glioblastoma (GBM) is hyper-vascularization. Bone marrow-derived macrophages are actively recruited to the tumor and referred to as glioma-associated macrophages (GAMs) which are thought to provide a critical role in tumor neo-vascularization. However, the mechanisms by which GAMs regulate endothelial cells (ECs) in the process of tumor vascularization and response to anti-angiogenic therapy (AATx) is not well-understood. Here we show that GBM cells secrete IL-8 and CCL2 which stimulate GAMs to produce TNF α . Subsequently, TNF α induces a distinct gene expression signature of activated ECs including VCAM-1, ICAM-1, CXCL5, and CXCL10. Inhibition of TNF α blocks GAM-induced EC activation both in vitro and in vivo and improve survival in mouse glioma models. Importantly we show that high TNF α expression predicts worse response to Bevacizumab in GBM patients. We further demonstrated in mouse model that treatment with B20.4.1.1, the mouse analog of Bevacizumab, increased macrophage recruitment to the tumor area and correlated with upregulated TNF α expression in GAMs and increased EC activation, which may be responsible for the failure of AATx in GBMs. These results suggest TNF α is a novel therapeutic that may reverse resistance to AATx. Future

clinical studies should be aimed at inhibiting TNF α as a concurrent therapy in GBMs.
